HIGHER TIER ONLY QUESTIONS

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)			$CO_2 + C \rightarrow 2CO$ award (1) for both reactants award (1) for product **and** balancing	2			2		
	(b)	(i)		$\boxed{2}\,Fe_2O_3 + 3C \longrightarrow \boxed{4}\,Fe + \boxed{3}\,CO_2$		1		1		
		(ii)		iron oxide is reduced ⇨ it loses oxygen (1) neutral answer – iron is reduced ⇨ it loses oxygen carbon is oxidised ⇨ it gains oxygen (1) if no other credit award (1) for generic description or for identifying which reactants are oxidised and reduced • oxidation is gaining oxygen, reduction is losing oxygen • iron oxide is reduced (to form iron) and carbon is oxidised (to form carbon dioxide) • oxygen goes from iron oxide to carbon reference to electrons is neutral		2		2		
	(c)	(i)		decomposition	1			1		1
		(ii)		$CaSiO_3$	1			1		
				Question 2 total	**4**	**3**	**0**	**7**	**0**	**1**